

# RAGHAV AWASTHI

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## EDUCATION

### University College London (UCL)

*Bachelor of Science in Computer Science, First Class Honours*

London, UK

Sep 2023 - Jun 2026

### Dubai College

*A\*A\*A\*A\*A in A-Levels and 11 grade 9s in GCSEs*

Dubai, UAE

Aug 2016 - Jun 2023

## EXPERIENCE

### AI/ML Engineering Intern

Jul 2026 - Present

*VotalAI*

- Implemented three configurable guardrail-serving modes - Votal, NVIDIA Nemotron and dual-model - on a feature branch by extending vLLM serving and the synchronous/asynchronous model router with concurrent verdict aggregation; kept 1,066 offline tests green and added two live API cases.

### Cybersecurity Analyst

Jun 2025 - Aug 2025

*Barclays*

*Knutsford, UK*

- Cut certificate triage time 25% and flagged 30+ high-risk certificates by building production Python services that enriched CA chains, classified root causes, standardised ServiceNow records and scored risk.

### Full-Stack Developer

Aug 2024 - Oct 2024

*Adamas Data*

*London, UK*

- Kept median query latency below 3 seconds across 140M+ rows and 150+ filter combinations by indexing PostgreSQL and PostGIS queries behind 30+ versioned REST endpoints.
- Enabled deployment rollback in under 10 minutes by containerising and automating CI/CD for a production FastAPI, PostgreSQL and Next.js application shipped end-to-end.

### AI/ML and Quantum Security Intern

Jun 2024 - Aug 2024

*Tech Mahindra*

*Pune, India*

- Raised domain F1 15% on a 6k-question benchmark and cut mean GPU inference latency 20% by leading a five-person team to LoRA fine-tune a 7B-parameter model on 120k curated records.

## PROJECTS

### Walrus-Cache-CDN (UCL Dissertation) | *Rust, RocksDB, Tokio, Sui, Prometheus*

Jun 2025 - Present

- Increased retrieval throughput 4.2x and cut average latency from 85.0 ms to 20.5 ms on a reproducible 100k-request benchmark by building a lifecycle-aware two-tier Rust cache for Walrus decentralised storage.
- Achieved an 81.7% hit rate by combining an in-memory hot tier, persistent RocksDB storage, W-TinyLFU/LRU admission and Sui event-driven invalidation with Prometheus observability.

### Quantum Proximity Gateway (IBM and UCL) | *Python, Rust, C++, Next.js*

Oct 2024 - May 2025

- Automated hot-desk PC authentication by integrating ESP32 BLE proximity, Raspberry Pi facial recognition and Pico USB-HID emulation into a distributed edge-to-cloud system.

## TECHNICAL SKILLS

**Languages:** Rust, Python, C/C++, Java, JavaScript/TypeScript, SQL, Bash

**Backend/Systems:** Distributed systems, REST APIs, microservices, caching, FastAPI, Node.js, React/Next.js

**Data/Storage:** RocksDB, PostgreSQL, PostGIS, Redis, SQLite, SQLAlchemy

**Infrastructure:** Tokio, Docker, Linux, CI/CD, Prometheus, Git, testing

## OPEN SOURCE AND LEADERSHIP

- Delivered merged contributions to Microsoft Terminal, mactop and Docusaurus, including a keyboard-driven Go help view and documentation navigation improvements.
- Served as UCL cricket vice-captain and first-year programming tutor; previously led the Dubai Computing Community and captained or coached cricket and volleyball teams.